

What is antimicrobial resistance?

- Antimicrobial resistance (AMR) is one of the biggest threats to both human and animal health. AMR can affect anyone, of any age, and in any country. AMR occurs when germs that can cause infections become resistant to medicines, such as antibiotics, that are used to kill them.
- Antimicrobials are medicines that kill or slow the growth of germs (bacteria, virus, fungus) that cause diseases. Antimicrobial resistance may lead to ‘treatment failure’, the inability to treat the cause of an infection. This can result in longer hospital stays, higher medical costs and even death.
- More information on AMR development is in this paper. [Meeting the Threat of Antibiotic Resistance: Building a New Frontline Defence](#).

What are the issues of AMR?

- AMR can occur naturally in germs but can also develop when people or animals take antimicrobial medicines. The germs become resistant to the medicines and continue to grow instead of dying. These resistant germs can spread and may infect other people or animals.
- In animals, AMR infections can reduce animal health, welfare and production outcomes. Sometimes resistant germs can spread from these animals to people who are in contact with them. Food from animals with resistant germs may also pose a risk to people if not stored, handled and cooked properly.
- We need to reduce the risk of germs developing resistance by:
 - only using microbials when needed
 - using antimicrobials appropriately
 - preventing infection in the first place.

Important resources

Australia’s Animal Sector Antimicrobial Resistance Action Plan 2023 to 2028

- [Australia’s Animal Sector AMR Action Plan 2023 to 2028](#) was released in September 2023.
- The plan includes activities from the [One Health Master Action Plan](#) and builds on the animal sector’s extensive work to address AMR. The plan supports [Australia’s National Antimicrobial Resistance Strategy – 2020 and Beyond](#).

AMR Vet Collective

- The [AMR Vet Collective](#) website has resources on best-practice use of antimicrobials in animals. There are engaging videos, decision trees, guidelines and online learning modules. Veterinarians can earn continuing education points for every module.

Good news in AMR surveillance

- The department works with major livestock industries to assess the levels of AMR in animals. This includes studies in the pork, chicken meat and egg, barramundi, and salmon industries. The level of AMR was either low or negligible against medicines that are important for humans.

Source: <https://www.agriculture.gov.au/agriculture-land/animal/health/amr>